Week 09: Scheduling: Critical Path and PERT

R&D and Project Management in Computer Science

R&D and Project Management in Computer Science (11117BLG002). Prof. Dr. Utku Kose. Full Professor, Department of Computer Engineering, Süleyman Demirel University, Isparta, Türkiye. Additional affiliations: University of North Dakota (USA), Universidad Panamericana (Mexico City, Mexico), Vel Tech University (Chennai, India). ORCID 0000-0002-9652-6415. E-mail: utkukose@sdu.edu.tr, utku.kose@und.edu, ukose@up.edu.mx, utkukose@gmail.com. This course is updated in line with current developments in the field. First version: September 2026. Last update: October 2026.

Overview

Once tasks and dependencies are known, the schedule follows from arithmetic. This week introduces network diagrams, the critical path method and PERT, and discusses how schedules can be compressed and where their assumptions fail [1, 2, 3].

Learning outcomes

By the end of the week, students are expected to draw a dependency network and to compute earliest and latest start and finish times and total float. They should be able to identify the critical path and to compute the probability of meeting a deadline with PERT. Finally, they should be able to explain crashing, fast tracking and the limits of PERT.

Lecture

Networks of tasks

A schedule begins with dependencies. In an activity-on-node network, each task is a node and an arrow from one task to another means that the second depends on the first. The most common dependency is finish-to-start, in which a task can start only when its predecessor has finished; start-to-start, finish-to-finish and start-to-finish dependencies describe other relations. In research projects, dependencies often run through approvals and data: Experiments cannot start before data access is granted, and a user study cannot start before ethics approval.

The critical path method

Kelley and Walker described the critical path method for planning and scheduling projects [1]. A forward pass through the network computes the earliest start and finish of every task: A task starts as soon as all its predecessors have finished. A backward pass from the project end computes the latest start and finish that do not delay the project. The difference between latest and earliest start is the total float, the delay a task can absorb without delaying the project. Tasks with zero float form the critical path, the longest path through the network, whose length is the

project duration. Figure 9.1 shows the schedule of a small research project with two critical paths that merge before the experiments.

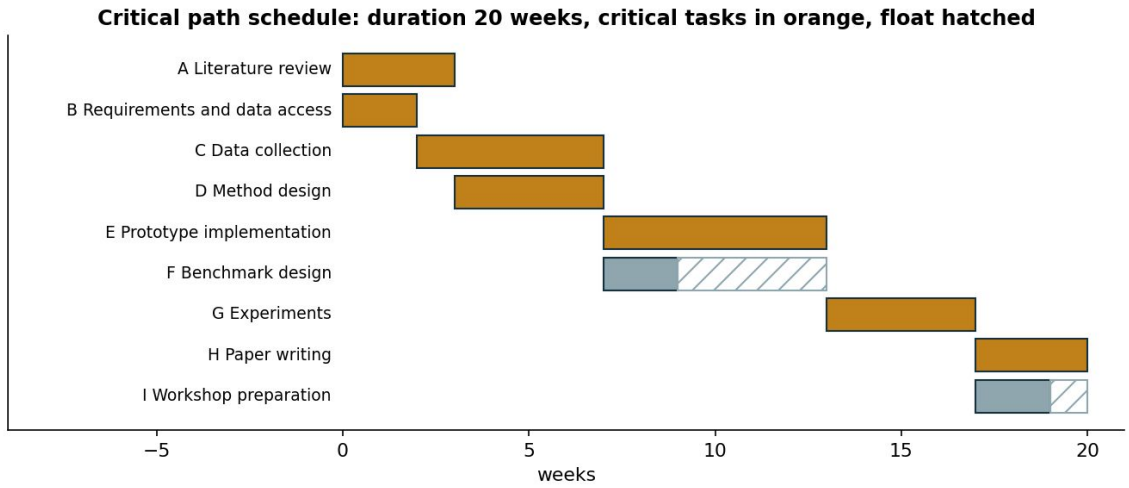


Figure 9.1. Critical path schedule of the example research project. Two critical paths of 20 weeks merge at the prototype, and benchmark design and workshop preparation have float [1].

PERT and the probability of meeting a deadline

PERT was developed for the research and development programme of the Polaris missile system [2]. It combines the network with three-point estimates. The expected duration of the project is the sum of the expected durations on the critical path. If tasks are independent, its variance is the sum of their variances. With a normal approximation, the probability of finishing by a deadline follows. The method has a known weakness: It considers only one critical path, so it ignores the chance that a near-critical path becomes critical. When paths merge, the expected completion time is later than PERT suggests. Week 10 addresses this merge bias with Monte Carlo simulation.

Compressing a schedule

When the computed duration exceeds the time available, the schedule can be compressed. Crashing adds resources to critical tasks to shorten them, at extra cost; fast tracking overlaps tasks that were planned in sequence, at extra risk [3]. Shortening a non-critical task gains nothing, and shortening a critical task can make another path critical. In research projects, the most effective compression is often to start long administrative tasks, such as data agreements and ethics applications, as early as possible.

> ***Pause and reflect.*** *Which task of your project lies on the critical path, and what could you start earlier to protect it?*

Hands-on components

Interactive lab: Edit the task list, one task per line: identifier, name, duration in weeks and predecessors separated by spaces. The calculator performs the forward and backward passes,

marks the critical path and estimates the probability of meeting a deadline with PERT [1, 2]. Link: https://utkukose.github.io/rd-project-management-NB-lecture/weeks/week-09/lab.html

Colab notebook: It implements the critical path method, computes the float of every task and estimates the probability of meeting a deadline with PERT [1, 2]. Link: https://colab.research.google.com/github/utkukose/rd-project-management-NB-lecture/blob/main/weeks/week-09/NB09_cpm_pert.ipynb

Reflection prompts

1. Enter your own project in the calculator. Which task on the critical path is most uncertain?
2. Where in your project could fast tracking create more risk than it saves time?
3. Why do administrative tasks so often lie on the critical path of research projects?

Weekly task and submission

What to do. Build the dependency network of your project from the tasks of Week 8. First, compute the critical path and the float of every task with the interactive lab or the Colab notebook. Second, add the schedule figure to your proposal draft. Third, state the probability of meeting the planned end date under PERT, and explain why this probability is likely to be too optimistic.

Where to look. Lecture: sections "The critical path method" and "PERT and the probability of meeting a deadline". Interactive lab: the critical path calculator, where you type your own task list (tab: Explore). Colab notebook: Section 1 with Exercise 1 for the critical path and Section 2 with Exercise 2 for the PERT probability.

Example. Suppose the critical path has an expected length of 40 weeks with a standard deviation of 3 weeks, and the planned end is week 44. Under PERT, the probability of finishing on time is about 91 percent. A second path that is almost as long makes the true probability lower.

What to submit. The network and the schedule figure with a short explanation of about 300 words, and the Colab notebook with Exercises 1 and 2 completed.

The weekly task supports self-learning and builds a personal portfolio. In an active semester, students who follow the course with the instructor can send their work for evaluation to utkukose@sdu.edu.tr or utkukose@gmail.com. The e-mail contains the items that the task lists under What to submit. It also includes the learning log exported from the interactive lab (tab: Reflect and submit).

References

[1] Kelley, J. E., & Walker, M. R. (1959). Critical-path planning and scheduling. In Proceedings of the Eastern Joint Computer Conference (pp. 160-173). https://doi.org/10.1145/1460299.1460318

[2] Malcolm, D. G., Roseboom, J. H., Clark, C. E., & Fazar, W. (1959). Application of a technique for research and development program evaluation. Operations Research, 7(5), 646-669. https://doi.org/10.1287/opre.7.5.646

[3] Project Management Institute (2021). A Guide to the Project Management Body of Knowledge (PMBOK Guide) and The Standard for Project Management (7th ed.). Project Management Institute.